ADI CN0549

The reference instrument — and two assumptions the measurement refuted

Chain	TE 8911 IEPE → CN0540 → AD7768-1
Converter	24-bit, DC–40 kHz
Carrier	Cora Z7-07S, Kuiper Linux, IIO
Power	3.0 W mains — IEPE needs 96 mW just to bias
Cost	∼EUR 700 against the STWIN's EUR 90
Role	**metrological, not operational**

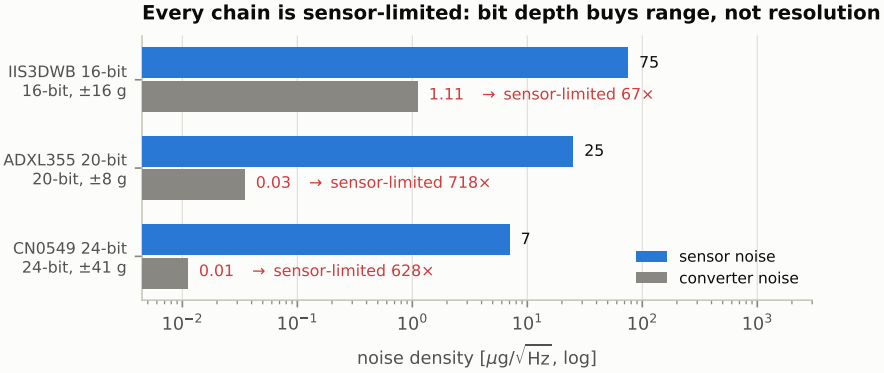
Every other kit's numbers rest on a datasheet. This kit's rest on a certificate: a serial number, a reference shaker and a date.

Its job is to be the instrument a fleet of EUR 90 nodes is compared against — deployed in ones, not hundreds.

Never run on hardware. Both findings below are simulator results, and both contradicted the design.

Branch `hw/cn0549-piezo-ae`. 15 kit tests, 36 shared edge-agent tests. Figures generated from `results/experiments.json`.

Finding 1: the 24-bit converter buys nothing



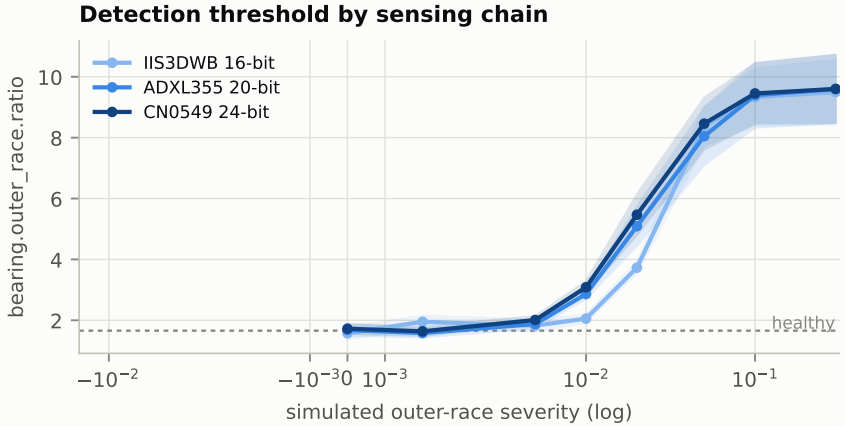
The first version of this experiment quantised one record to 16, 20 and 24 bits and compared the BPFO harmonic ratio. **All three agreed to three decimal places** at every severity from 0.002 to 0.6.

That was not a bug. Every chain here is sensor-limited by 67–718×: adding converter bits to a sensor-limited measurement buys dynamic range, not resolution.

A 16-bit MEMS part's ADC is *not* what limits it. Its own 75 $\mu g/\sqrt{\text{Hz}}$ sensor noise is 67× above its quantisation floor.

The same arithmetic appears in the agent's start-up report for every channel, and is asserted as a test — if it ever fails, the chain became converter-limited and this slide needs rewriting.

And a 10× quieter sensor did not detect earlier either



All three chains first separate fault from healthy at simulated severity **0.01**. The piezo gives a stronger signature once one exists (3.26 against 2.00) — useful for *severity estimation*, not for detection.

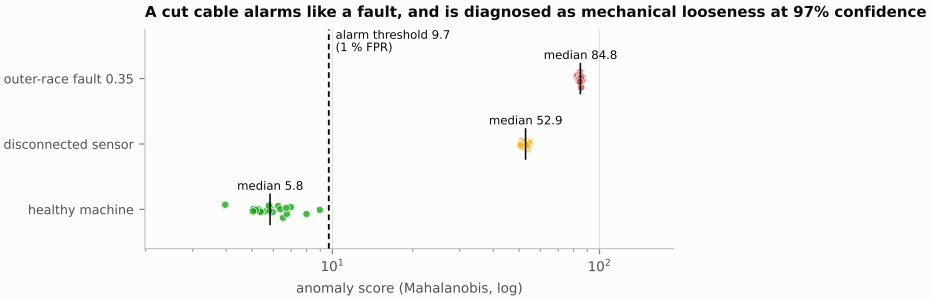
The machine's own broadband noise, not the sensor's, sets the floor.

So why buy it? Not resolution.

Bandwidth: DC–40 kHz against the IIS3DWB's 6.3 kHz. Lubrication distress emits at 20–40 kHz, which no MEMS part here can see *at all*. Categorical, not incremental.

Traceability: when a fleet disagrees with itself, a certificate settles it.

Finding 2: a cut cable is diagnosed as a fault



The prediction was that converter noise — the smoothest signal the system will ever see — would score *below* healthy, making the worst possible condition invisible.

Wrong. Distance from healthy is not ordered by smoothness: converter noise is far outside the healthy manifold, in a different direction, and a quadratic-form detector does not care which. Featureless is as anomalous as faulty.

The real failure is a level up.
The detector alarms, and the rule base then diagnoses the noise: **mechanical looseness, 97 % confidence**.

A fitter is dispatched to a healthy machine; the real fault is a connector; and when the fitter finds nothing, the programme loses credibility in the way that ends programmes.

Unfixable in the data. It needs an out-of-band check.

The out-of-band check: IEPE bias

An IEPE sensor idles at a known bias on its constant-current supply, so the bias voltage *is* the cable's state.

Bias	State	
$\geq$ 85 % of supply	`open`	cut cable
6–16 V	`ok`	live sensor
$\leq$ 1 V	`short`	shorted line
anything else	`unknown`	not a pass

A pure function of (voltage, supply rail), so the whole decision table is tested with no hardware — which matters, because this is the check that catches the failure the data cannot show.

`open()` **refuses to start, it does not warn.**

The costs are asymmetric. A node that refuses produces a loud, immediate, correctly-attributed failure. A node that starts on a cut cable produces months of data showing a beautifully smooth machine — and **nobody questions good news**.

`unknown` is not a pass either. That band is where a wrong sensor type, an over-long cable and a failing current source live, and "probably fine" is how the previous slide begins.

Open item: the bias should be *trended*, not just gated. A slowly falling bias is a failing source or a wetting connector, visible weeks before it trips a threshold.

So what: The most valuable feature on a EUR 700 instrument is a voltmeter reading on its own cable.

Two configuration choices that silently change the numbers

The anti-alias filter is a choice, not a default.

	Transition	Group delay
`sinc5`	sharp	long
`wideband`	flat to 0.433×	short

Spectra want sinc5. **Impacts and envelopes want wideband**, because sinc5's long impulse response smears the impacts the envelope detector exists to find.

So `filter_type` has no default and the agent prints it at start.

The clip check comes before DC removal.

This chain is DC-coupled, so the bias appears as a large offset that must be removed before any spectral work — left in, it leaks out of bin 0 into the low orders through the window skirts.

But the converter clips on the *total* signal including the offset. Removing it first and then checking would hide exactly the condition being looked for: a signal riding a large offset that saturates on one side only.

The removed amount is reported, not discarded — a drifting DC offset tracks the current source's health and the sensor's temperature.

What is missing, and what it would settle

IEPE bias decision table	5 tests
Config + calibration validated	6 tests
Experiments reproducible	JSON committed
Shared agent end-to-end	36 tests
Sensor-limited claim	asserted as a test
Run against a CN0549	no
IIO attribute names verified	no
Reference-shaker calibration	no
Cross-validation vs MEMS	no
20–40 kHz AE band	no
Calibration certificate	placeholder

The highest-value missing experiment in this repository: mount a CN0549 channel and a MEMS node on the *same* bearing housing, run a seeded-defect rig, and compare velocity RMS, kurtogram band choice, BPFO/BPFI ratios and detection severity.

That replaces finding 1's simulator result with a real one — and it is the only way to know whether a EUR 90 node's numbers can be trusted.

Second: 20–40 kHz under deliberate lubrication starvation. It is the kit's strongest remaining argument and nothing here has measured it.

So what: A bandwidth-and-traceability purchase wearing a resolution label. Buy it to validate a fleet, not to be one.